CCNA 200-301 V2.0

16 Static Routing, IPv4 and IPv6

Part III — IP Routing

EXAM TOPICS

3.2

LECTURER

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THE COURSE

Where this chapter sits

Part I	Foundations	chapters 1–7
Part II	Switching and Network Access	chapters 8–14
Part III	**IP Routing**	chapters 15–19
Part IV	Network Services and Security	chapters 20–26
Part V	Wireless, Virtualisation and Operations	chapters 27–32
Part VI	Sitting the Exam	chapters 33–34

LEARNING OUTCOMES

What you will be able to do

- **Configure** the four kinds of static route the blueprint names: default, network, host and floating.
- **Choose** correctly between a next-hop address, an exit interface, and both — a choice that matters more than it looks.
- **Configure** static routing for IPv6 alongside IPv4.
- **Troubleshoot** a static route that is absent from the table, or present and not working.

ASSUMED KNOWLEDGE

Before we start

Chapter 15 — longest prefix match and administrative distance are assumed throughout. Chapter 5 for the v6 half.

KEY TERMS

Vocabulary for this chapter

- static route
- default route
- network route
- host route
- floating static
- next hop
- exit interface
- fully specified route
- recursive lookup
- gateway of last resort

Each is defined where it first matters — not here.

ON THE EXAM

The verb is troubleshoot, and the sub-topics are a checklist

This is what the blueprint actually asks for

Topic **3.2** is “troubleshoot IPv4 and IPv6 static routing” and then lists exactly four kinds: default route, network route, host route, floating static. Expect to be shown a table and a symptom rather than asked to recite syntax — but you cannot diagnose what you cannot write, so learn the syntax first.

The four kinds

CONFIGURE

All four, on IPv4

IOS · configuration mode

```
! Network route -- reach 10.9.9.0/24 through the neighbour at 10.255.0.2.
ip route 10.9.9.0 255.255.255.0 10.255.0.2
!
! Default route -- the gateway of last resort.
ip route 0.0.0.0 0.0.0.0 203.0.113.1
!
! Host route -- this one server goes a different way.
ip route 10.9.9.50 255.255.255.255 10.255.1.2
!
! Floating static -- distance 200 keeps it out of the table while
! OSPF (distance 110) has a route. It appears only if OSPF loses one.
ip route 10.1.0.0 255.255.0.0 10.255.2.2 200
```

CONCEPT

Why the host route wins

All four routes above can match `10.9.9.50`. The `/32` host route is the longest prefix, so it is used — ahead of the `/24`, ahead of the default. Longest prefix match decides before distance is even considered (Chapter 15).

Next hop, exit interface, or both

WATCH OUT

Exit-interface-only routes on Ethernet are a trap

Ethernet is multi-access: many devices share the segment. A route that names only an exit interface tells the router “send it out here” without saying to whom, so the router must ARP for *every destination address* it forwards, hoping something answers. On a link where proxy ARP is enabled it appears to work; where it is not, it silently fails; and either way the ARP table fills with entries that should not exist.

On a serial point-to-point link there is only one possible recipient, so the form is harmless. On Ethernet, always give the next hop — alone or fully specified.

VERIFY

show ip route static

```
R1#

       10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
S         10.1.0.0/16 [200/0] via 10.255.2.2
S         10.9.9.0/24 [1/0] via 10.255.0.2
S         10.9.9.50/32 [1/0] via 10.255.1.2
S*     0.0.0.0/0 [1/0] via 203.0.113.1
```

Static routing for IPv6

CONFIGURE

IPv6 static routes

IOS · configuration mode

```
ipv6 unicast-routing
!
! Network route via a global next-hop address.
ipv6 route 2001:db8:acad:20::/64 2001:db8:acad:99::2
!
! Default route.
ipv6 route ::/0 2001:db8:acad:99::2
!
! Host route.
ipv6 route 2001:db8:acad:20::50/128 2001:db8:acad:99::2
!
! Floating static.
ipv6 route 2001:db8:acad:30::/64 2001:db8:acad:98::2 200
!
! Via a LINK-LOCAL next hop -- the exit interface is mandatory,
! because fe80:: addresses are only unique per link and the router
! has no way to know which interface you mean.
ipv6 route 2001:db8:acad:40::/64 GigabitEthernet0/1 fe80::2
```

CONCEPT

Why link-local next hops need the interface

A link-local address is valid on one link only, and the same `fe80::2` may legitimately exist on three different interfaces of the same router. Naming the interface removes the ambiguity. IOS rejects a link-local next hop without one.

This matters because routing protocols, including OSPFv3, use link-local addresses as their next hop — so link-local next hops are the normal case in IPv6, not an oddity.

VERIFY

show ipv6 route static

R1#

```
IPv6 Routing Table - default - 7 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
S   ::/0 [1/0]
     via 2001:DB8:ACAD:99::2
S   2001:DB8:ACAD:20::/64 [1/0]
     via 2001:DB8:ACAD:99::2
S   2001:DB8:ACAD:40::/64 [1/0]
     via FE80::2, GigabitEthernet0/1
```

Floating statics, properly

WORKED EXAMPLE

A backup path that only activates on failure

`R1` reaches `10.1.0.0/16` through OSPF over its primary link, and has a slow secondary link to the same destination.

- OSPF's route has distance 110.
- Configure the backup with a distance *above* 110: `ip route 10.1.0.0 255.255.0.0 10.255.2.2 200`.
- While OSPF is up, only the OSPF route is installed. `show ip route` shows `O 10.1.0.0/16 [110/20]` and no static entry at all.
- The primary link fails. OSPF withdraws its route. The static route, now the only candidate, is installed as `S 10.1.0.0/16 [200/0]`.

WORKED EXAMPLE

A backup path that only activates on failure (continued)

- The primary returns. OSPF’s route at 110 beats 200 and displaces the static again.

The number 200 is not magic — it just has to exceed the distance of whatever it is backing up. Back up a *static* route (distance 1) and any value from 2 upwards will do.

Common pitfalls

- **A floating static with too low a distance.** Set it below the dynamic protocol’s and it takes over permanently, which is the opposite of a backup.
- **Exit-interface-only routes on Ethernet.** See the warning above.
- **Forgetting the reverse route.** A static route is one-directional. Both ends need one, and forgetting this is the commonest static-routing fault of all.
- **A next hop that is not directly reachable.** The route will not be installed at all; the router silently discards it. `show ip route` simply has no entry, which looks like you never typed it.
- **Omitting the interface on an IPv6 link-local next hop.** Rejected outright.

The static route was typed and it is not in the table.

An engineer configures `ip route 10.9.9.0 255.255.255.0 10.255.0.2` and `show ip route static` shows nothing. No error was displayed when the command was entered.

The static route was typed and it is not in the table.

What would you check first — and what do you expect to see?

show ip route 10.255.0.2

```
R1#
% Network not in table
```

show ip interface brief | include Gigabit

```
R1#
GigabitEthernet0/1     10.255.0.1      YES manual administratively down down
```

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. A static route is only installed if the router can resolve its next hop. `10.255.0.2` is not reachable — there is no connected subnet covering it — so the route is accepted into the configuration and never enters the routing table. This is why the command produced no error: the syntax was perfect and the topology was not.

Check the interface that should reach it:

And the lesson

Fix. The interface is shut. `no shutdown` on `Gi0/1`, and the connected route for `10.255.0.0/30` appears, the next hop resolves, and the static route installs itself immediately. Confirm with `show ip route static` — and remember the general rule: **a static route missing from the table is nearly always an unreachable next hop**, not a typing error.

Four routes, one failure

Topology. Three routers — `R1`, `R2`, `R3` — with `R1` connected to `R2` by a fast link and to `R3` by a slow one, and `R2` and `R3` both connected to a LAN at the far side. A LAN on `R1`.

1. Address everything. Confirm connected routes on all three and that neighbours ping.
2. On `R1`, add a network route to the far LAN via `R2`. Test from the `R1` LAN — it will fail. Diagnose why, then add the reverse route and re-test.
3. Add a default route on `R2` and `R3` pointing back at `R1`. Note the `S*` and the gateway of last resort line.
4. Add a host route on `R1` for one server in the far LAN, via `R3`. Prove with `show ip route <address>` that it is preferred over the network route.
5. Add a floating static via `R3` for the far LAN with distance 5. Confirm it is *not* in the table.

Four routes, one failure (continued)

6. Shut R1's link to R2. Confirm the floating static appears and traffic still flows. Restore the link and watch it withdraw.
7. **Extension.** Repeat steps 2 and 5 for IPv6, including one route via a link-local next hop.

You configure a static route and it never appears in the table, with no error message. What is the most likely cause and which command confirms it?

Discuss · then advance

Checkpoint 1 of 3

You configure a static route and it never appears in the table, with no error message. What is the most likely cause and which command confirms it?

> The next hop is not reachable. A router will not install a static route whose next-hop address it cannot resolve to an interface. Confirm with `show ip route <next-hop>` — if that returns nothing, the static route has nowhere to point.

A floating static backing up an OSPF route is configured with distance 100. What will happen, and why is it wrong?

Discuss · then advance

ANSWER

Checkpoint 2 of 3

A floating static backing up an OSPF route is configured with distance 100. What will happen, and why is it wrong?

> Nothing will happen: 100 is *lower* than OSPF's 110, so the "backup" route wins permanently and the OSPF route is never installed. A floating static must have a distance *higher* than the protocol it backs up.

Why must an IPv6 static route using a link-local next hop also name an exit interface?

Discuss · then advance

ANSWER

Checkpoint 3 of 3

Why must an IPv6 static route using a link-local next hop also name an exit interface?

> Because a link-local address is only unique within a link, and a router with several interfaces has no way to tell which one `fe80::2` refers to. Naming the exit interface removes the ambiguity.

What to take away

- Four kinds: network, default (`0.0.0.0/0`), host (`/32`) and floating. The host route wins on longest prefix match.
- Give a next hop on Ethernet — alone or fully specified. Exit-interface only belongs on point-to-point links.
- IPv6 mirrors IPv4, but a link-local next hop *must* name the exit interface.
- A floating static needs a distance above whatever it backs up: over 110 for OSPF, over 1 for another static.
- A static route absent from the table almost always means an unresolvable next hop. Routes are one-way; configure both ends.

Where we got to

- Four kinds: network, default (`0.0.0.0/0`), host (`/32`) and floating. The host route wins on longest prefix match.
- Give a next hop on Ethernet — alone or fully specified. Exit-interface only belongs on point-to-point links.
- IPv6 mirrors IPv4, but a link-local next hop *must* name the exit interface.
- A floating static needs a distance above whatever it backs up: over 110 for OSPF, over 1 for another static.

NEXT

Chapter 17 — OSPFv2, Single Area